

PROJECT

Module

Data Acquisition
and Processing

Module Code

AI53009FP

Duration

16 hours

Title: Perform data collection and preprocessing tasks

Tools, Equipment, and Materials:

1. Computer with Internet access
2. Python 3.x with Jupyter Notebook
3. Presentation software (PowerPoint or Google Slides)

Scenario: Singapore Tourism Data Analysis and Prediction

This project requires you to demonstrate your skills in data collection from APIs, data integration, preprocessing, feature engineering, and building a predictive model using real-world data from the Singapore Tourism Board and hotel statistics from data.gov.sg.

Project Task:

1. This is an individual project. All work must be your own.
2. The project consists of five parts covering data collection, data cleaning, preprocessing, feature engineering, and modelling.
3. You may refer to online documentation (e.g., pandas, scikit-learn, requests) for syntax help.
4. No copying of code wholesale is allowed. Code must be written and understood by you.
5. Document your code with comments explaining your approach.

Deliverables:

Submit a zipped folder containing the following:

- Jupyter notebooks (.ipynb) - containing all your Python code with outputs and markdown explanations.
- Presentation slides (.pptx) - summarising your project process and findings.
- Exported data files - cleaned and processed datasets in CSV format.

Note: Ensure only one submission per student. Name your zip file as:

StudentIndex_Name_DAP_Project.zip

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PROJECT SCENARIO

Background

The Singapore Tourism Board (STB) wants to understand patterns in tourism arrivals and predict future visitor volumes. As a data analyst, you are tasked with collecting data from Singapore's open data APIs, integrating multiple data sources, and building a predictive model to forecast tourism trends.

Problem Statement

"Predict visitor arrival patterns and identify key factors influencing tourism trends using historical tourism data."

Data Sources

You will collect data from the following Singapore Government Open Data APIs (data.gov.sg):

- **Primary Dataset:** International Visitor Arrivals to Singapore (Monthly)
- **Secondary Dataset:** Hotel Statistics (for data integration)

API Documentation: <https://data.gov.sg/datasets>

PART 1: DATA COLLECTION

Tasks

1. **Set up your environment**
2. **Extract data from the primary API**
3. **Extract data from the secondary API**
4. **Data Integration**
5. **Initial Data Inspection**

PART 2: DATA CLEANING AND VALIDATION

Tasks

1. **Identify Data Quality Issues**
2. **Handle Missing Values**
3. **Remove Duplicates**
4. **Detect and Handle Outliers**
5. **Data Validation**
6. **Export Cleaned Data**

PART 3: DATA PREPROCESSING

Tasks

1. **Data Transformation – Encoding Categorical Variables**
2. **Train-Test Split**
3. **Feature Scaling**
4. **Verify Preprocessing**

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PART 4: FEATURE ENGINEERING

Tasks

1. **Create New Features**
2. **Feature Selection using Correlation Analysis**
3. **Dimensionality Reduction**
4. **Justify Selections**

PART 5: MODELLING AND EVALUATION

Tasks

1. **Model Training**
2. **Model Evaluation**
3. **Model Comparison – save as an image file**
4. **Best Model Analysis**
5. **Model Prediction and Visualisation**

PRESENTATION REQUIREMENTS

Your presentation should summarise your project process and findings. The content should be organised according to the following in this sequence:

1. **Cover Slide:** Project title, student ID, and your name.
2. **Problem Statement:** Define the problem using the 4Ws framework (Who, What, Where, Why).
3. **Data Sources:** Describe the APIs used and data collected.
4. **Data Quality:** Summarise data quality issues found and how you addressed them.
5. **Feature Engineering:** Explain the new features created and the feature selection process.
6. **Model Results:** Present model comparison and performance metrics.
7. **Key Insights:** Share the most important findings and feature importances.
8. **Conclusion:** Summarise your approach and suggest improvements.

Important Notes

- **Academic Integrity:** Plagiarism or copying code will result in zero marks. All work must be original.
- **Technical Issues:** If you encounter API access issues, inform your lecturer immediately for alternative data sources.
- **Help Resources:** You may refer to official documentation (pandas, scikit-learn, requests), but must write your own code.

- END -